1. Project Title: Development of a Dynamic Ride-Sharing and Carpooling Platform

2. Project Statement and Outcomes:

The "Smart Ride Sharing System" project aims to develop a website-based platform where vehicle owners traveling long distances can share their available seats with passengers heading in the same direction. This system will allow drivers to post their trips and passengers to book seats based on their destinations. The fare calculation will be dynamic, depending on the distance each passenger travels.

The outcome will be a user-friendly web application that facilitates seamless ride-sharing. It will include features like route-based matching, fare estimation, secure payment integration, and user reviews. The project will ensure transparency, cost-effectiveness, and sustainability by reducing the number of vehicles on the road while promoting shared mobility.

3. Modules to be Implemented:

- User Management Module
- Ride Posting and Booking Module
- Fare Calculation & Payment Module
- Route Matching & Tracking Module
- Notification and Review System
- Admin Dashboard for Monitoring

4. Week-wise Module Implementation and High-Level Requirements:

Weeks 1-2: Management & Ride Posting Module

- Implement user registration and authentication for drivers and passengers.
- Store user profiles, including vehicle details for drivers.
- Secure password storage and login authentication.
- Enable drivers to post available rides with source, destination, date, and seat availability.
- Allow passengers to search for rides based on their destination and book seats.

- Implement a booking confirmation system with ride details.

Weeks 3-4: Fare Calculation, Payment & Route Matching Module:

- Implement dynamic fare calculation based on distance traveled.
- Integrate online payment options.
- Ensure transaction history is stored for users.
- Ensure real-time updates for both drivers and passengers

Weeks 5-6: Notification System, Review, and Admin Dashboard

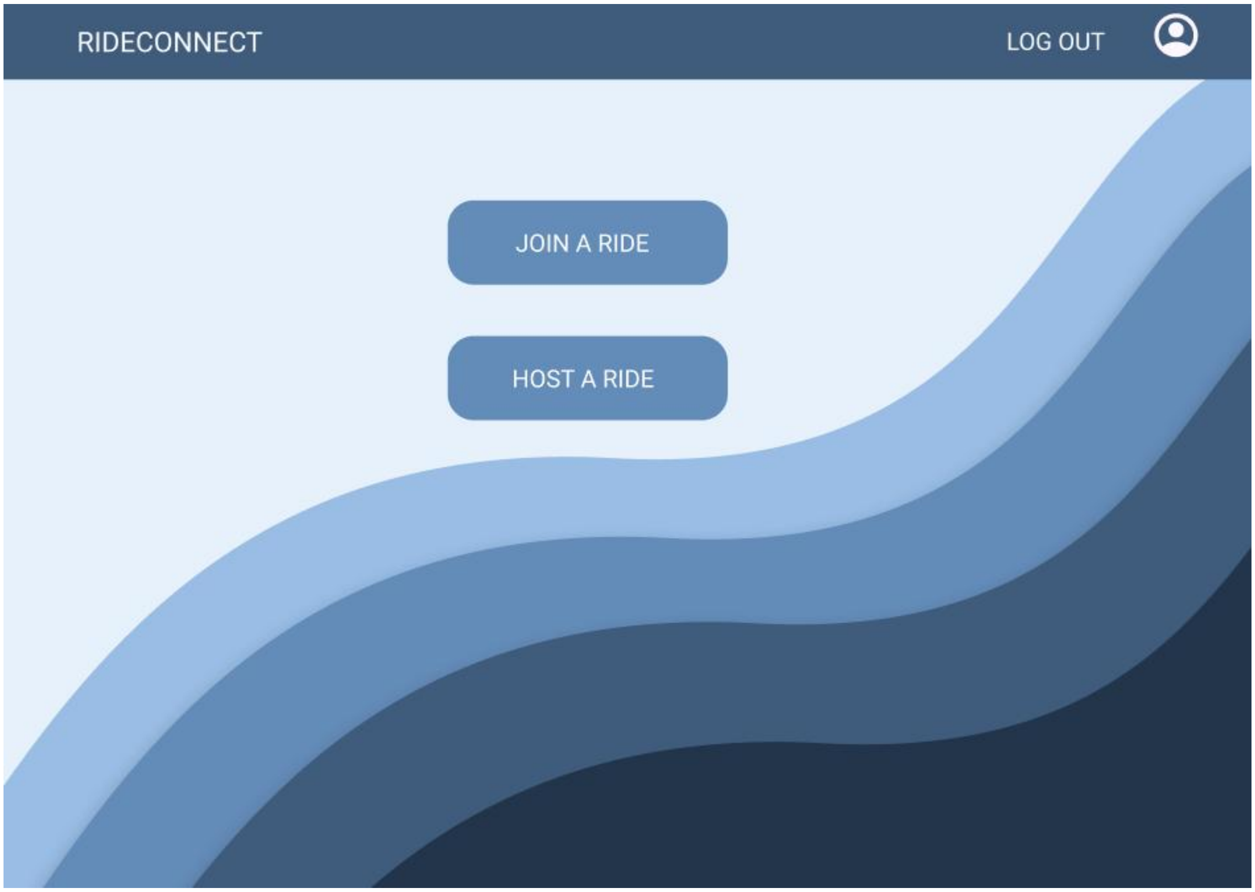
- Implement an in-app notification system for ride confirmations, reminders, and updates.
- Allow passengers and drivers to review each other after the ride.
- Develop an admin panel to monitor rides, payments, and user activities.

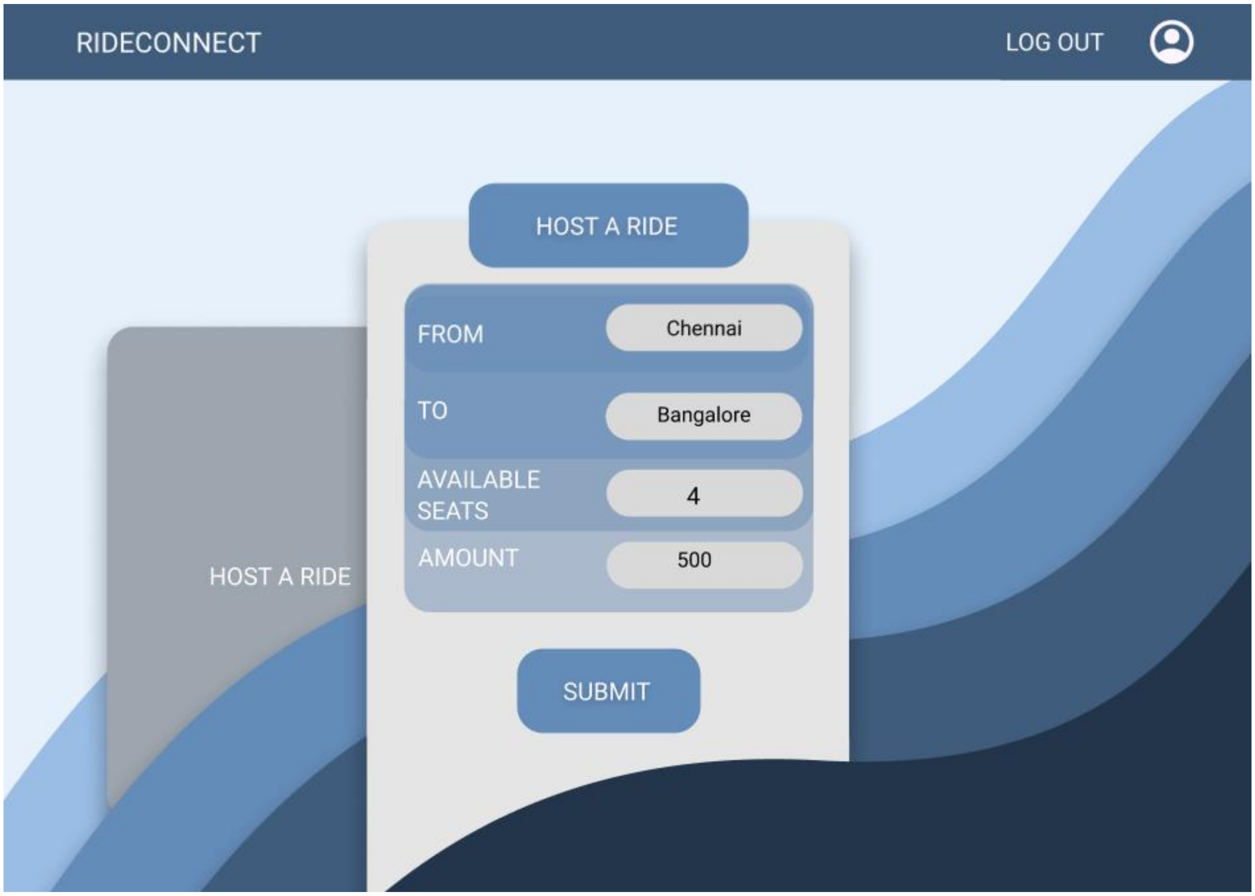
Weeks 7-8: Testing, Review, and Documentation

- Conduct extensive testing to ensure the platform's stability, usability, and security.
- Perform unit testing, integration testing, and user acceptance testing.
- Finalize and compile technical and user documentation.
- Prepare a deployment guide and troubleshooting documentation.
- Complete project review and make final adjustments.

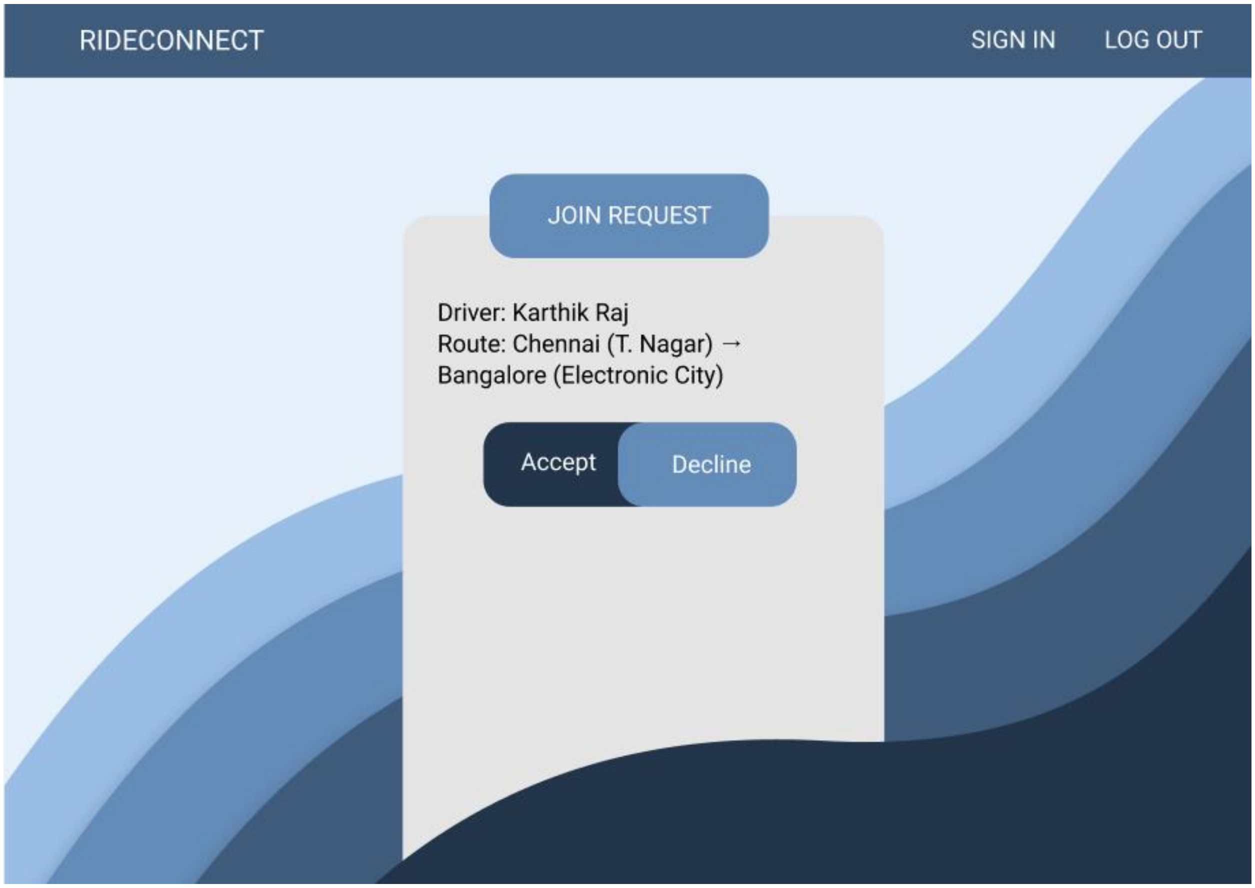
5. Flow chart:

6. Sample Dashboard:


RIDECONNECT
LOG OUT
JOIN A RIDE
HOST A RIDE


RIDECONNECT
LOG OUT
HOST A RIDE
HOST A RIDE
FROM
Chennai
TO
Bangalore
AVAILABLE SEATS
4
AMOUNT
500
SUBMIT


RIDECONNECT
SIGN IN
LOG OUT
JOIN A RIDE
JOIN
SEARCH AVAILABILITY RIDES
AVAILABLE RIDES
Driver: Karthik Raj
Departure: 7:00 AM, Feb 27, 2025
Pickup Location: T. Nagar, Chennai
Drop-off Location: Electronic City, Bangalore
Available Seats: 4
Fare: ₹750 per seat
JOIN
Driver: Priya Sharma
Departure: 5:30 PM, Feb 28, 2025
Pickup Location: Adyar, Chennai
Drop-off Location: Majestic, Bangalore
Available Seats: 2
Fare: ₹800 per seat
JOIN


RIDECONNECT
SIGN IN
LOG OUT
JOIN REQUEST
Driver: Karthik Raj
Route: Chennai (T. Nagar) → Bangalore (Electronic City)
Accept
Decline